

STEGANOGRAPHY TOOL

(Report)

Submitted By

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Project Title

Steganography Tool

Academic Year

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1. Introduction

Steganography is the technique of hiding secret information inside a digital medium such as an image, audio, or video file. Unlike cryptography, which encrypts data, steganography conceals the existence of the message itself.

In this project, a simple steganography tool is developed using Python in Kali Linux. The tool allows users to hide (encode) and retrieve (decode) secret messages within an image file.

2. Objective

To understand the concept of steganography and data hiding.

To develop a Python-based tool for hiding secret messages in images.

To implement encoding and decoding techniques using image pixels.

To learn the use of Python libraries like PIL (Pillow).

To gain practical knowledge of cybersecurity concepts.

3. Tools & Technologies Used

Operating System: Kali Linux

Programming Language: Python 3

Library Used: PIL (Python Imaging Library / Pillow)

Editor: Nano / Terminal

4. Methodology

The project works in two main parts:

◆ **Encoding** (Hiding Data)

The user enters a secret message.

The program converts the message into binary format.

It modifies the least significant bits (LSB) of image pixels.

A new image (output.png) is created with hidden data.

◆ **Decoding** (Retrieving Data)

The program reads pixel values from the image.

Extracts the hidden binary data.

Converts it back to the original message, Displays the secret message.

5. Steps to Execute the Project

Install required library:

```
sudo apt install python3-pil -y
```

Place files in same folder:

steganography.py

input.png

Run the program:

```
python3 steganography.py
```

Choose option:

1 → Encode

2 → Decode

Follow instructions to hide or retrieve message.

6. Sample Output

Encoding:

Input Image: input.png

Secret Message: "Hello"

Output Image: output.png

Decoding:

Input: output.png

Output: "Hello"

7. Advantages

Provides secure way to hide information

Simple and easy to use

Useful in cybersecurity applications

Maintains image quality

8. Limitations

Works best only with PNG images

Limited message size

Can be detected with advanced analysis

9. Applications

Secure communication

Digital watermarking

Data protection

Military and intelligence use

10. Conclusion

The steganography tool successfully demonstrates how secret data can be hidden and retrieved from images using Python. This project helps in understanding data security concepts and practical implementation of steganography techniques.

Output ScreenShot:

```
kali@kali: ~  
$ sudo apt install python3-pil -y  
[sudo] password for kali:  
Upgrading:  
  libavif16  python3-pil  python3-pil.imagetk  
Installing dependencies:  
  libsvtav1enc4  
Summary:  
  Upgrading: 3, Installing: 1, Removing: 0, Not Upgrading: 1645  
  Download size: 3,202 kB  
  Space needed: 7,348 kB / 62.0 GB available  
Get:1 http://http.kali.org/kali kali-rolling/main amd64 libsvtav1enc4 amd64 4.1.0+dfsg-1 [2,414 kB]  
Get:2 http://kali.download/kali kali-rolling/main amd64 python3-pil imagetk amd64 12.2.0-1 [63.4 kB]  
Get:3 http://kali.download/kali kali-rolling/main amd64 python3-pil amd64 12.2.0-1 [558 kB]  
Get:4 http://http.kali.org/kali kali-rolling/main amd64 libavif16 amd64 1.4.1-1+b1 [147 kB]  
Fetched 3,202 kB in 11s (276 kB/s)  
Selecting previously unselected package libsvtav1enc4:amd64.  
(Reading database ... 40965 files and directories currently installed.)  
Preparing to unpack ./libsvtav1enc4_4.1.0+dfsg-1_amd64.deb.  
Unpacking libsvtav1enc4:amd64 (4.1.0+dfsg-1).  
Preparing to unpack ./libavif16_1.4.1-1+b1_amd64.deb.  
Unpacking libavif16:amd64 (1.4.1-1+b1) over (1.3.0-1+b2).  
Preparing to unpack ./python3-pil-imagetk_12.2.0-1_amd64.deb.  
Unpacking python3-pil-imagetk:amd64 (12.2.0-1) over (12.0.0-1+b1).  
Preparing to unpack ./python3-pil_12.2.0-1_amd64.deb.  
Unpacking python3-pil:amd64 (12.2.0-1) over (12.0.0-1+b1).  
Setting up libsvtav1enc4:amd64 (4.1.0+dfsg-1).  
Setting up libavif16:amd64 (1.4.1-1+b1).  
Setting up python3-pil:amd64 (12.2.0-1).  
Setting up python3-pil-imagetk:amd64 (12.2.0-1).  
Processing triggers for libc-bin (2.42-13)...  
kali@kali: ~  
$
```

```
kali@kali: ~/Downloads  
$ sudo apt install python3-pil -y  
[sudo] password for kali:  
python3-pil is already the newest version (12.2.0-1).  
Summary:  
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 1645  
kali@kali: ~  
$ nano steganography.py  
1. Encode  
2. Decode  
Enter your choice: 1  
Traceback (most recent call last):  
  File "/home/kali/steganography.py", line 51, in <module>  
    encode()  
  File "/home/kali/steganography.py", line 6, in encode  
    img = Image.open('input.png')  
  File "/usr/lib/python3/dist-packages/PIL/image.py", line 3635, in open  
    fp = builtins.open(filename, "rb")  
FileNotFoundError: [Errno 2] No such file or directory: 'input.png'  
kali@kali: ~  
$ cd Downloads  
kali@kali: ~/Downloads  
$ ls  
input.png  
kali@kali: ~/Downloads  
$ mv  
/home/kali/Downloads  
kali@kali: ~/Downloads  
$ mv ~/steganography.py  
mv: missing destination file operand after '/home/kali/steganography.py'  
Try 'mv -h' for more information.  
kali@kali: ~/Downloads  
$ mv ~/steganography.py  
kali@kali: ~/Downloads  
$ ls  
input.png  steganography.py  
kali@kali: ~/Downloads  
$ python3 steganography.py  
1. Encode  
2. Decode  
Enter your choice: 1  
Enter secret message: Hello World  
Message encoded successfully!  
kali@kali: ~/Downloads  
$
```